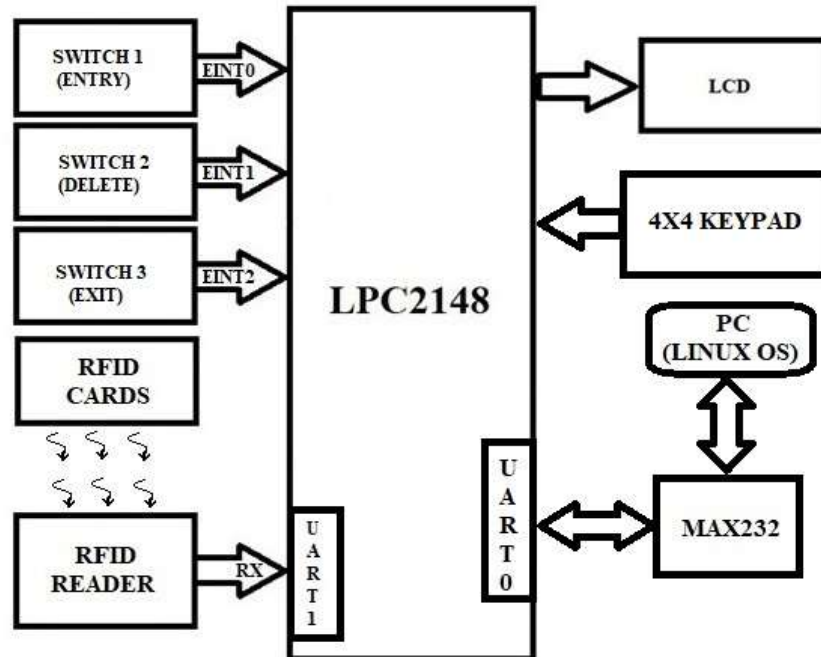


SMARTCART: RFID SMART BILLING SYSTEM

AIM:

The main aim of the project is to develop the general billing system by using RFID technology and data base integration.

BLOCK DIAGRAM:



REQUIREMENTS:

HARDWARE REQUIREMENTS:

- LPC2148
- RFID READER
- RFID CARDS
- LCD
- SWITCHES
- 4X4 KEYPAD
- MAX232
- USB-TO-UART CONVERTER

SOFTWARE REQUIREMENTS:

- EMBEDDED C – PROGRAMMING
- KEIL-C COMPILER
- FLASH MAGIC

Steps to be followed to complete your project:

- Create New Folder in your server save that folder with your project name.
- In this project two different application programs are required to communicate to get the final result. One application program is for microcontroller board and other one C program which is used for communicating the microcontroller board with UART communication with database integration from LINUX OS.
- Follow the below mentioned steps to complete the project.
- Copy what you do with files like lcd.c, lcd.h, delay.c, delay.h, uart.c, uart.h, keypad.c, keypad.h into project folder.
- Individually can check each and every module.
- First check lcd to display character constant, string constant and integer constant.
- Next check keypad peripheral by displaying key values on LCD.
- Next check uart0 and uart1 peripherals by transmitting character constant, string constant and receive string constant using hyper terminal. (Note: use UART interrupt concept for both the UART's (UART0 & UART1))

Note: By using UART0 driver code, you need to develop UART1 driver.

- Connect RFID READER D0 pin to USB to UART converter and test the working condition.
- Then write logic for reading the card number using UART1 and display it on LCD.
- Then the external interrupts (EINT0, EINT1 and EINT2) working condition on hardware.
- If above steps are completed create new file with projectmain.c, add all peripheral definition files, and write below steps in projectmain.c file
- Initially project name will display on LCD for some specific time and in continuous process, it is waiting for Entry key input from the user to access the entry process. If user is selected the entry option by pressing Entry key (EINT0), the application program is waiting for RFID cards (Manager card or items card). During this time, it is accessing the delete process by selecting the Delete key or Exit process by selecting the Exit key. If Manager card placed, then this card number will send to the PC through UART0 in specific format (MCARDNUMBERS). In Linux system, once it receives the manager card then application program is giving support for updating the new stock information into the database based on user input. If item related card is placed, then this card number will send to the PC through UART0 in specific format (CCARDNUMBERS). In Linux system, once it is received the item related card it will read the card number equivalent cost from the data base and it will reduce the quantity from the database then it will send the items remaining quantity along with cost to the microcontroller board through UART0. Microcontroller board will receive

the data and item cost were added to the total billing amount. This process will continue till it receives the signal from Exit key/Delete key. If signal received from delete key, then same process will continue like Entry key process but in reverse/opposite way. In this process, deleting message format (**DCARDNUMBERS**) needs to be used. If Exit key equivalent signal is received, then one menu will display for the payment option.

- Payment process is as follows. Menu contains 1. Cash 2. Card. User has to select the required option from the keypad. If cash option is selected, then user has to enter the bill amount from the keypad. If success, then payment process completed and again application program is waiting for Entry key signal. If card option is selected, then application program is waiting for ATM card. If the ATM card is placed then send the atm card number along with billing amount in one specified format (**BCARDNUMBERABILLAMOUNTS**). Then it is waiting for the PIN conformation from the C code. If it receives the signal for PIN, then user has to enter the ATM PIN from the keypad and it will send to the Linux PC for validation. After validation if it receives the success response the payment process is done otherwise it will display the corresponding message. For wrong PIN, it attempts for three times. If three times failed, then payment process failed. If payment process failed then all items quantity required to update to the data base.
- LINUX CODE WORKING: First download the `UART_LOOPBACK_CODE_FOR_LINUX` code from LMS card under reference programs and check the working of that code on Linux system.
- Initially create two different text files. One is used to maintain the stock information second one is used to maintain the bank accounts information. After executing the Linux code executable code, initially it will display the all-stocks information on terminal and waiting for the card information from the microcontroller board. Based on the card it received, it will proceed accordingly. Once refer the microcontroller board working then it will give you the clear picture for this Linux code implementation.
- If you're getting this output then your project is completed.

Note: if same application is completed for more than one bank account, it is more effective/impressive.

Text files information. Just for reference

Stock.txt:

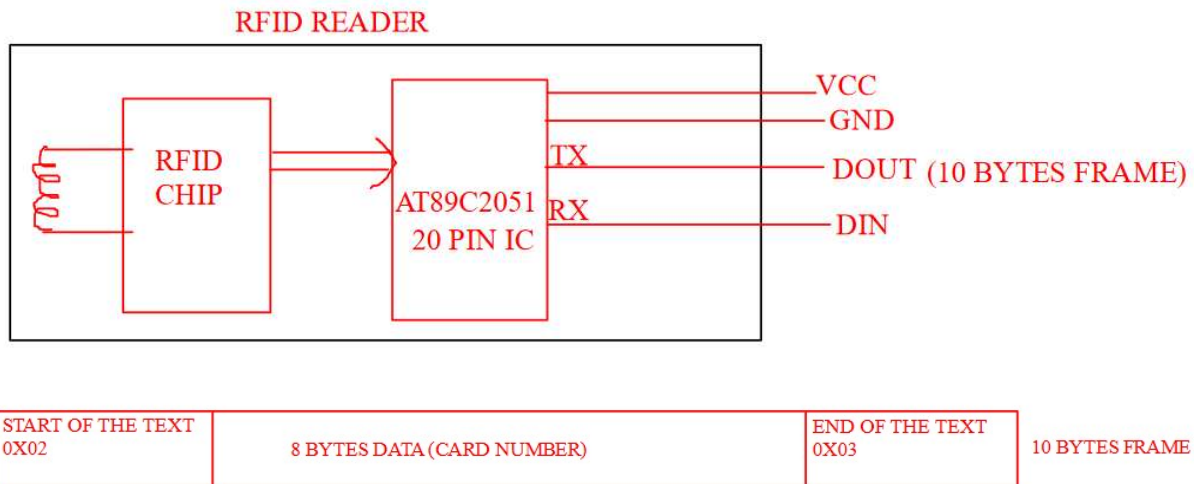
Item name	Card Number	Quantity	Price
SOAPS	00332069	190	100
MILKY	00336463	95	50
CHIPS	00312472	26	10

Bank.txt:

PLACE	IFSC CODE	BANK NAME	ACCOUNT NUMBER	BAL	PIN
HYDERABAD	099V23HE9	VECTOR	12638754	10000	1234

*****ALL THE BEST*****

RFID READER: RFID Reader will send the 10 bytes of data once card is placed nearer to the RFID Reader. Once check the below images to get some idea on RFID Reader block diagram. Refer the supported data sheets to get some more knowledge.



For example, card number is 12345678, then output of the RFID Reader is **0x02 0x31 0x32 0x33 0x34 0x35 0x36 0x37 0x38 0x03** (hex format)